Data Visualization

Study Plan & Resources

Everything you need to learn Matplotlib & Seaborn — from basic plots to full visual storytelling.

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Study Topics

Part 1

Core Visualization Topics

The fundamental concepts every student must know before building real charts and dashboards..

Topic	What to Know
Matplotlib Basics	figure(), axes, plt.plot() — understanding the Figure/Axes object model and the functional vs OOP API.
Seaborn Integration	sns.set_style(), load_dataset(), and how Seaborn wraps Matplotlib to give you statistical chart types with less code.
Line & Scatter Plots	lineplot(), scatterplot(), and plt.plot() — encoding trends and relationships with markers, colors, and hue.
Distribution Plots	histplot(), kdeplot(), ecdfplot() — reading the shape, spread, and skew of a variable's distribution.
Bar & Count Plots	barplot() and countplot() — comparing categories with aggregated values or raw frequencies.
Box & Violin Plots	boxplot() and violinplot() — comparing distributions across groups and reading medians, IQR, and outliers.
Heatmaps	sns.heatmap() with pivoted DataFrames — visualizing correlation matrices and time patterns in a grid.
Subplots & OOP API	plt.subplots(), the fig/ax pattern, and tight_layout() — composing multiple chart types in one figure, using axes with seaborn.
Styling & Themes	plt.style.use(), sns.set_style(), palette, and color arguments — applying consistent, professional visual style.
Customization	titles, axis labels, ticks, legends, and grid — every chart needs these before it communicates anything.

Part 2

Important Use Cases

How the chart types above connect to real EDA, reporting, and communication workflows.

Once you know the tools, you need to know when and why to use them. These are the key workflows that matter in practice:

Exploratory Data Analysis (EDA)

Using scatter plots, histograms, and heatmaps to visually profile a dataset before drawing any conclusions — understanding shape, outliers, and relationships at a glance.

Visual Storytelling

Choosing the right chart for each data type, then using color, titles, and layout to make the chart communicate a finding — not just display numbers.

Distribution Analysis

Using histplot, kdeplot, and violinplot together to fully characterize how a variable is distributed across the dataset or across groups.

Comparison Across Groups

Using hue, boxplots, and bar charts to expose meaningful differences between categories — smoker vs non-smoker, species, day of week, and so on.

Time Series & Heatmaps

Using lineplot and heatmap with pivoted DataFrames to reveal trends, seasonality, and patterns over time — the flights dataset is the perfect example.

Dashboard Composition

Organizing multiple chart types into a coherent subplot grid using the OOP API — so your analysis reads as a structured visual report, not a pile of disconnected figures.

Part 3

Resources

Curated materials to help you learn and go deeper — start with the team's own content first.

From the AI Sub-Team

VIDEO **Data Visualization with Matplotlib & Seaborn**
Our team's own session — start here before going anywhere else.
[Session]

Notebook
Our team's Python Notebook (the same one written in the video, with extra formatting and comments).
[NoteBook]

NOTE: download and open from Vscode for better formatting.

Playlists & External Guides

Use the documentation, it's a great reference.

Matplotlib
A complete walkthroughs of Matplotlib from scratch — great if you want to understand the library top to bottom.
Playlist by Corey
[Matplotlib Tutorial --- Corey Schafer]

Video by BroCode
[**Learn Matplotlib in 1 hour!** 📊]

PLAYLIST **Playlist 2**
Seaborn in one focused session — covers every chart type used in the task with real datasets.

Video by Derek Banas --- Great over all
[**Seaborn Tutorial : Seaborn Full Course**]

Video by NeuralNine – Shorter but covers similar topics to what we covered (Recommended for a quick review)
[**Seaborn Crash Course - Data Visualization in Python**]